PROJECT DOCUMENTATION: REAL-TIME MOTION-TRIGGERED SECURITY AND EDGE ANALYSIS SYSTEM

Group Members:

1. **Curato, Althea Faith J.** **Section:** FNHX1
2. **Garong, Jia Marie M.** **Section:** EUHX1
3. **Reyes, Jamaya Althea** **Section:** FNHX1

1. Title and Objectives of the Project

- **Project Title:** Real-Time Motion-Triggered Security and Edge Analysis System
- **Project Objectives:**
    - To design and implement a working computer vision application using Python and OpenCV that integrates classical image processing and analysis techniques to solve a real-world problem.
    - To apply core laboratory concepts including Image Operations, Image Segmentation, and Image Analysis/Transformation.
    - To visualize key detection results cleanly using bounding boxes, labels, and real-time status overlays.

2. Description of the Problem or Use-Case

In modern residential, institutional, and commercial environments, automated monitoring is essential for security and asset oversight. Conventional surveillance setups often require expensive hardware or heavy machine learning dependencies.

This project addresses the real-world problem of lightweight, low-latency area monitoring and motion-triggered security. The application acts as an automated digital sentry that processes a live webcam stream in real-time. It remains in a passive monitoring state when an area is secure, and automatically flags and tracks foreground motion with bounding boxes the moment an unauthorized activity or movement occurs within the frame.

3. Techniques Used (OpenCV Topics)

The application successfully integrates three core classical computer vision techniques from the laboratory modules:

- **a. Image Operations:**

Raw webcam frames undergo foundational preprocessing. Each captured frame is resized for performance efficiency, converted from BGR to **Grayscale** to reduce matrix dimensionality, and passed through a **Gaussian Blur** filter (cv2.GaussianBlur) to eliminate high-frequency sensor noise and minor lighting fluctuations.

- **b. Image Segmentation:**

To isolate moving subjects from the static background, the system implements **Frame Differencing** and **Thresholding** (cv2.absdiff and cv2.threshold). By comparing the current grayscale frame with a reference background frame, pixel differences are thresholded into a binary mask. **Contour Extraction** (cv2.findContours) is then applied to detect and group the contours of moving objects.

- **d. Image Analysis and Transformation:**

The system employs **Canny Edge Detection** (cv2.Canny) on the processed grayscale stream. This structural analysis technique computes image gradients to extract object boundaries and fine edges. The resulting edge map is rendered live inside a picture-in-picture preview box on the main display interface for real-time visual analysis.

4. Screenshots of Results and Output

Figure 1: *Application running in normal state displaying "Status: Monitoring..." and the live Canny Edge analysis preview box in the top-right corner.*

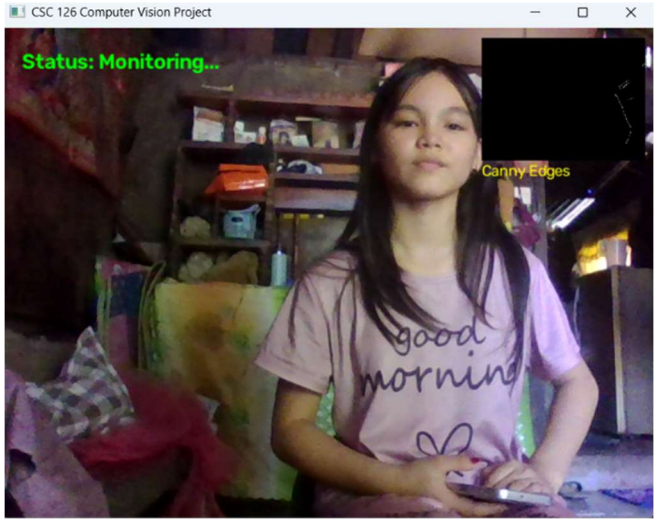
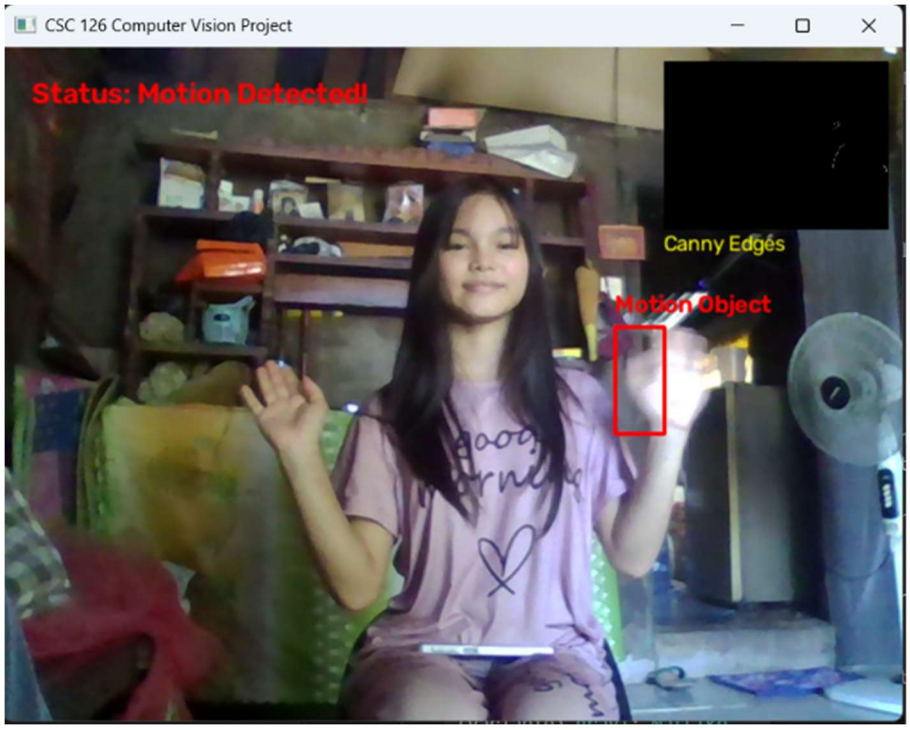


Figure 2: *Application reacting to movement, displaying "Status: Motion Detected!" in red and rendering a bounding box over the tracked moving object.*



5. Discussion of Accuracy, Behavior, and Limitations

- **Behavior:** The application exhibits stable behavior under consistent indoor lighting. When the scene is static, the difference matrix remains low, maintaining the "Monitoring" state. Sudden pixel intensity shifts instantly trip the motion threshold, drawing visual bounding boxes and updating status alerts.
- **Accuracy:** The application achieves high accuracy for high-contrast, distinct foreground movements. Gaussian blurring effectively prevents false positives caused by minor sensor noise.
- **Limitations:**
    - The frame-differencing segmentation method is sensitive to abrupt global illumination changes (such as lights turning on/off or harsh shifting sunlight).
    - Very slow, gradual movements can sometimes fall below the threshold sensitivity, delaying detection.

6. Conclusion

The developed system successfully fulfills all functional and technical criteria outlined in the CSC 126 Midterm Project guidelines by delivering a working, real-time computer vision application using classical OpenCV principles. Through this project, we gained practical insights into how fundamental image operations, spatial transformations, and threshold-based segmentation can be combined to build effective automation tools without heavy machine learning frameworks. Future iterations could incorporate adaptive background subtraction algorithms (such as MOG2) to handle lighting variations more robustly and include automated frame-logging capabilities when security alerts are triggered.